
PROFESSIONAL SUMMARY

Software engineer focused on backend systems, distributed data pipelines, and cloud-native services. Experience building scalable microservices, designing high-throughput data flows, and improving system reliability across large engineering teams. Strong foundation in Go, Python, Java, JavaScript/TypeScript, and Kubernetes-based deployments. Skilled at driving architectural improvements, automating developer workflows, and enhancing application performance in production environments.

TECHNICAL SKILLS

Languages: Go, Python, Java, JavaScript, TypeScript, Bash, C

Backend & Frameworks: Maven, Gradle, SpringBoot, Node, React, React Native, Vue, Flask, FastAPI, NextJS

Cloud & DevOps: AWS S3, Docker, Kubernetes, Helm, Ansible, Openshift

Data & Streaming: SQL, KDB+, Pinot, Flink, Kafka, Polars, Pandas, Elastic Search

Testing & Monitoring: JUnit, Robot Framework, Cucumber, Mockito, Opentelemetry, Grafana

Design and architecture: Caching strategies, Database design, Data tracing, REST

EMPLOYMENT HISTORY

Data Engineer / Citigroup Inc.

Mississauga, Canada / July 2025 – Present

- Architected and delivered a high-throughput, S3-backed parquet data warehouse supporting large-scale algorithmic backtesting on **~1.4 TB of daily market data** for market traders and quants. Replaced the legacy pipeline with a parallelized, non-blocking ingestion and transformation architecture using Python multiprocessing and columnar processing (Polars, PyArrow), driving a **5× performance improvement** and significantly increasing system scalability.
- Designed and introduced a distributed tracing pipeline for Apache Flink real-time applications using OpenTelemetry, enabling end-to-end context propagation, data completeness, timeliness validation, and proactive troubleshooting of production issues—establishing observability capabilities previously absent within the team.
- Built and maintained high-throughput data pipelines for fixed income and commodities trading, leveraging Python, Kafka, and Flink to power both real-time and batch processing, with optimized storage and query layers in Pinot, Elasticsearch, and S3 to support low-latency analytics and trading use cases as part of day to day duties.

Software Engineer / Citigroup Inc.

Mississauga, Canada / July 2024 – July 2025

- Proactively identified gaps in cross-service observability and led the design and rollout of a unified database schema across multiple microservices and API gateways, **reducing error diagnosis time by 50%** and materially improving operational visibility across the engineering team.
- Took ownership of performance bottlenecks in a latency-sensitive client application, redesigning critical execution paths and introducing targeted caching strategies that **reduced API response times from 9s to 3s** across core user journeys.
- Drove automation of recurring code-quality and maintenance workflows by building a lightweight Bash tool integrated into all CI pipelines, **eliminating manual steps and saving ~96 engineering hours** annually.
- Strengthened service security posture by standardizing OAuth-based access control and SSL certificate management across all microservices, and delivered an internal authentication library adopted as the default across team products.

Mobile Application Tech Lead / iRespite Services

Montreal, Canada / January 2023 – December 2023

- Served as **Technical Lead** for the iRespite MHealth platform, owning end-to-end architecture, system design, implementation, and code review. Delivered a production-ready React Native application backed by Firebase infrastructure and 42 API endpoints, supporting three distinct user groups—patients, family members, and care providers.
- Drove technical vision and product execution from prototype through stakeholder presentation, **securing a \$108K startup grant** by demonstrating architectural soundness, scalability, and real-world clinical impact to investors and healthcare professionals.

EDUCATION

Bachelor of Engineering - McGill University, Montreal, Canada

Major: Computer Engineering | **Minor:** Mathematics

Honors and Awards: James McGill Scholarship, RCN Undergraduate Award

LEADERSHIP EXPERIENCE

McGill Electrical, Computer, and Software Engineering Student Society (ECSESS)

May 2021 - May 2022

Led as the **Vice-President of Internal Affairs** for a 1200+ member engineering student society. Advanced initiatives in networking, inclusivity, and sustainability. Significantly enhanced member engagement and fostered a more collaborative and accessible environment for everyone.